

Information Theory and Its Application in Deep Learning: Final

June 2026

- You can use previous parts of a problem even if you did not solve them.
- Logarithms are natural (nats) unless stated otherwise; for the rate-distortion problem use base 2 ($\log = \log_2$, rates in bits).
- $D(\cdot\|\cdot)$ denotes the KL divergence and $h(\cdot)$ the differential entropy.

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1	/ 30
2	/ 35
3	/ 35
Total	/ 100

extra sheet

1. **Differential Entropy and Maximum Entropy** (30 points)

Let X be a continuous random variable. Recall $h(X) = -\int f(x) \log f(x) dx$ (nats).

- (a) Let $X \sim \text{Exp}(\lambda)$, i.e. $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$. Compute $h(X)$. (10 points)
- (b) Let $Y = cX$ for a constant $c > 0$. Find $h(Y)$ in two ways: (i) from the scaling rule, and (ii) by identifying the distribution of Y . (10 points)
- (c) Show that among *all* densities supported on $(0, \infty)$ with mean $\mathbb{E}[X] = \mu$, the exponential density $f(x) = \frac{1}{\mu} e^{-x/\mu}$ maximizes h . (10 points)

2. **Rate–Distortion of an Independent Gaussian Pair** (35 points)

Consider a vector source $\mathbf{X} = (X_1, X_2)$ with *independent* components $X_i \sim \mathcal{N}(0, \sigma_i^2)$, and squared-error distortion $d(\mathbf{x}, \hat{\mathbf{x}}) = \sum_{i=1}^2 (x_i - \hat{x}_i)^2$. Work with the information rate–distortion function

$$R(D) = \min_{p(\hat{\mathbf{x}}|\mathbf{x}): \mathbb{E}\|\mathbf{X} - \hat{\mathbf{X}}\|^2 \leq D} I(\mathbf{X}; \hat{\mathbf{X}}),$$

and write R_i for the single-component function $R_i(D_i) = \frac{1}{2} \log \frac{\sigma_i^2}{D_i}$ ($0 < D_i \leq \sigma_i^2$). Rates are in bits, $\log = \log_2$. The goal is to prove

$$R(D) = \min_{\substack{D_1, D_2 \geq 0 \\ D_1 + D_2 \leq D}} (R_1(D_1) + R_2(D_2)).$$

(a) **Decoupling lemma.** For *any* joint distribution of $(\mathbf{X}, \hat{\mathbf{X}})$, show

$$I(\mathbf{X}; \hat{\mathbf{X}}) \geq I(X_1; \hat{X}_1) + I(X_2; \hat{X}_2).$$

Hint: write $I = h(\mathbf{X}) - h(\mathbf{X} | \hat{\mathbf{X}})$; use independence on the first term, the chain rule on the second, and “conditioning reduces entropy.” (10 points)

(b) **Converse.** Show that

$$R(D) \geq \min_{D_1 + D_2 \leq D} (R_1(D_1) + R_2(D_2)).$$

Hint: take any test channel feasible for $R(D)$; set $D_i = \mathbb{E}[(X_i - \hat{X}_i)^2]$ so $D_1 + D_2 \leq D$. Apply part (a), then bound each $I(X_i; \hat{X}_i)$ below using the definition of the single-letter R_i . (10 points)

(c) **Achievability.** Show the matching upper bound

$$R(D) \leq \min_{D_1 + D_2 \leq D} (R_1(D_1) + R_2(D_2)),$$

so that equality holds. *Hint: fix any split (D_1, D_2) with $D_1 + D_2 \leq D$ and build a product test channel $p(\hat{\mathbf{x}} | \mathbf{x}) = \prod_i p_i^*(\hat{x}_i | x_i)$ from the per-component optimizers. Compute its mutual information and its distortion. (15 points)*

3. **The χ^2 -Divergence: From Estimator Variance to the Evidence** (35 points)

The χ^2 -divergence is $\chi^2(P\|Q) = \int \frac{(p-q)^2}{q} = \int \frac{p^2}{q} - 1$. After establishing its basic properties, you will use it twice: once to lower-bound the variance of an estimator, and once to upper-bound an intractable log-evidence.

- (a) **Foundations.** Show $\chi^2(P\|Q) = \text{Var}_Q(\frac{dP}{dQ})$, and that $D(P\|Q) \leq \chi^2(P\|Q)$.
Hint: $\log t \leq t - 1$. (10 points)
- (b) **Hammersley–Chapman–Robbins.** For any statistic T with $\mathbb{E}_P[T] = a$, $\mathbb{E}_Q[T] = b$, prove

$$\text{Var}_Q(T) \geq \frac{(a - b)^2}{\chi^2(P\|Q)}.$$

Then, for a parametric family $\{P_\theta\}$ with an unbiased estimator of θ , take $Q = P_\theta$, $P = P_{\theta+h}$ and let $h \rightarrow 0$ to recover the Cramér–Rao bound $\text{Var}_\theta(T) \geq 1/I(\theta)$. *Hint: express $a - b$ as a covariance under Q and apply Cauchy–Schwarz. (15 points)*

- (c) **A χ^2 upper bound on the evidence (CUBO).** For a latent-variable model $p(x) = \int p(x, z) dz$, a proposal $q(z | x)$, and weight $w = \frac{p(x, z)}{q(z | x)}$, define $\text{CUBO}_2 = \frac{1}{2} \log \mathbb{E}_{q(z|x)}[w^2]$. Show $\log p(x) \leq \text{CUBO}_2$, that the gap is exactly $\frac{1}{2} \log(1 + \chi^2(p(z | x) \| q(z | x)))$, and hence $\text{ELBO} \leq \log p(x) \leq \text{CUBO}_2$. *Hint: $p(x) = \mathbb{E}_q[w]$ and $(\mathbb{E}_q[w])^2 \leq \mathbb{E}_q[w^2]$; then write $w = p(x) \frac{p(z|x)}{q(z|x)}$. (10 points)*

